



Workload Picked
Batch of requests through various Engines
LLM Request Timelines
NSYS Measurements
Throughput vs Load for vLLM
Future Steps

# Benchmarking and Performance Analysis of LLM Inferencing Systems

Systems for LLMs
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# Workload Picked





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- Targeted a very specific workload
- Large system prompt ($\approx$ 1500 tokens) containing useful context and shared across all the queries
- Varying query input rate (from 5 req/sec to 700 req/sec)
- Small query sizes (10 to 20 tokens each) and small output requirements (capped at 30 to 50 tokens)
- Ideally, prefix sharing should help a lot
- **Next Step:** Family of requests, each family with a particular large system prompt and a few queries

# Batch of requests through various Engines





| **Technique Used** | **Time (s)** |
|---|---|
| Transformers (batch size = 1) | 30.21 |
| Transformers (batch size = 2) | 19.51 |
| Transformers (batch size = 3) | 18.64 |
| vLLM (no system prompt pre-filling) | 0.45 |
| SGLang (fork) | 1.03 |
| SGLang (run_batch) | 0.40 |
| NEO | 0.55 |

Table: Measuring the time taken to process a batch of 24 requests using different techniques

# LLM Request Timelines

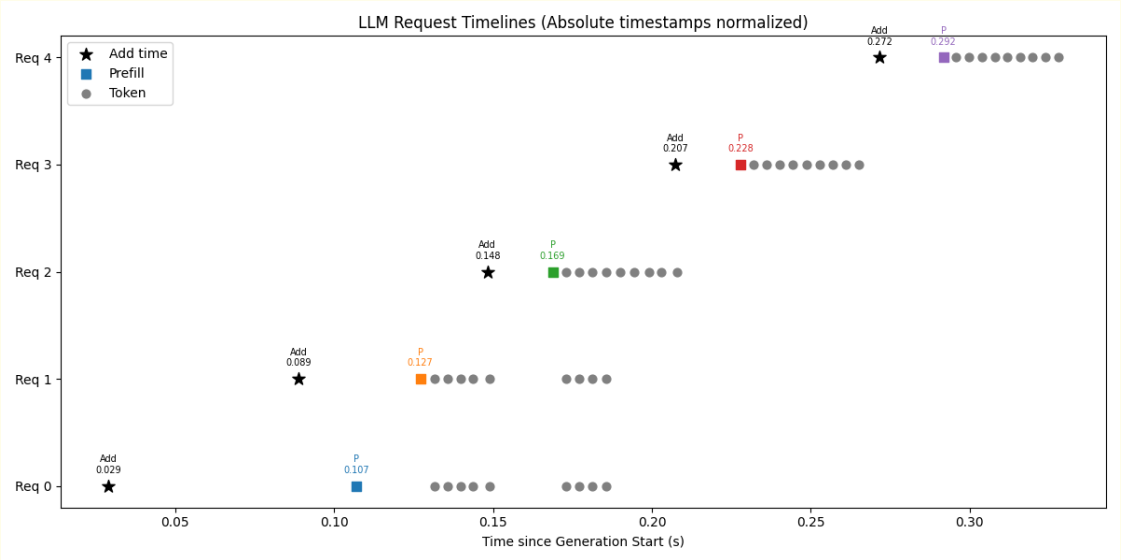



Figure: 20 Requests per second - without chunked prefill - vLLM V0 Engine

# NSYS Measurements




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# Throughput vs Load for vLLM

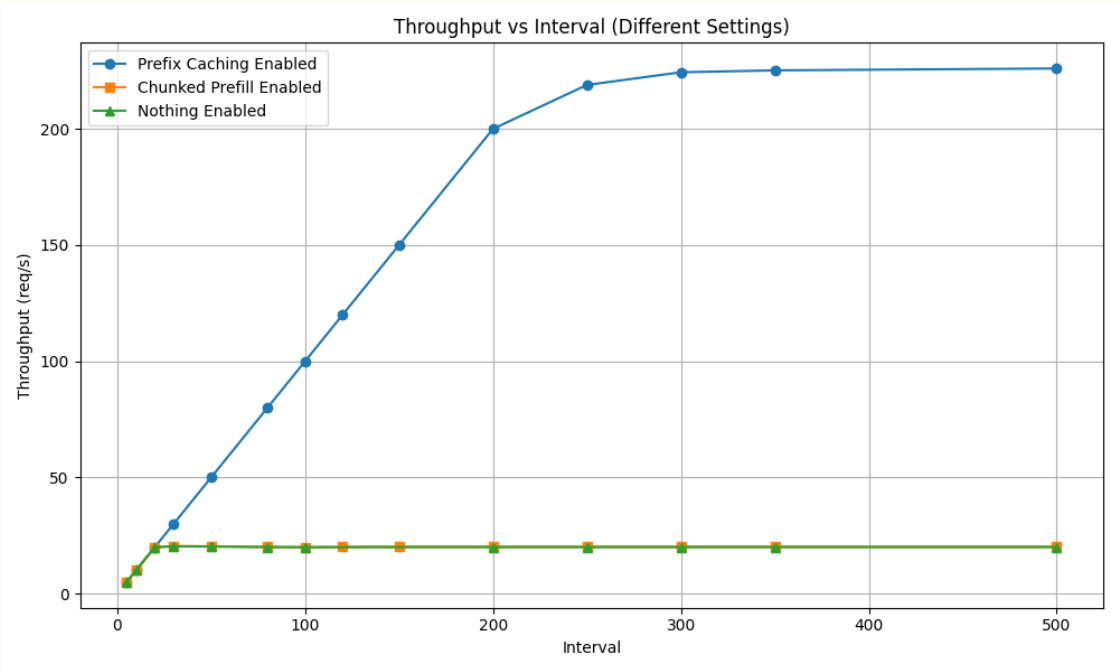



Figure: Throughput vs Load


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# Thank You